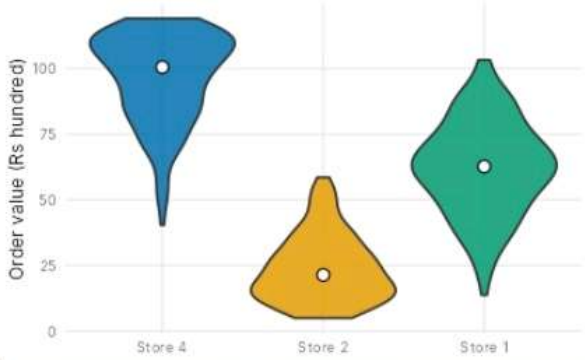
DSA06 - Data Handling and Visualization | Activity: Identify Skewness using Violin Plots | Slot C

R.No: 1 Reg. No: 192524157 Name: A Lokesh Kumar

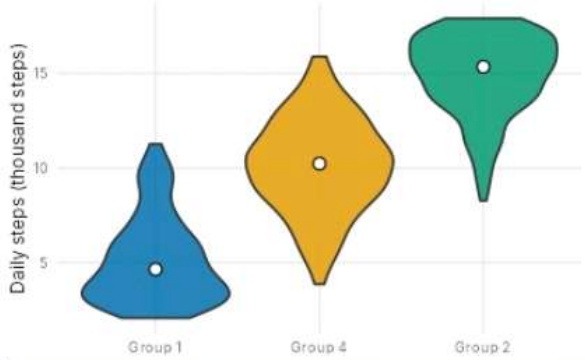
Part A (18 marks). Each chart is a violin plot (white dot = median) made in R with ggplot2. For every violin circle the skew (1 mark) and the relation of the mean to the median (0.5 mark).

Chart 1: Order value (Rs hundred)



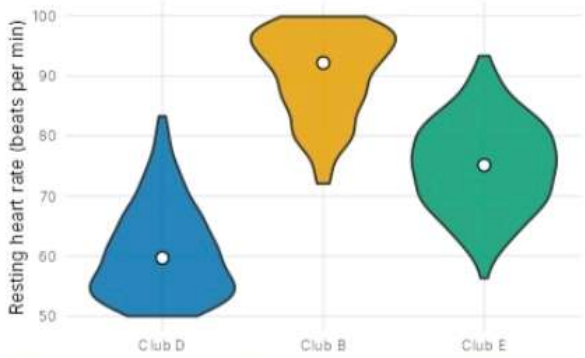
Violin	Skew (circle one)	Mean vs Median (circle)
Store 4	Right / Left / Symm.	Mean > / < / ~ Median
Store 2	Right / Left / Symm.	Mean > / < / ~ Median
Store 1	Right / Left / Symm.	Mean > / < / ~ Median

Chart 2: Daily steps (thousand steps)



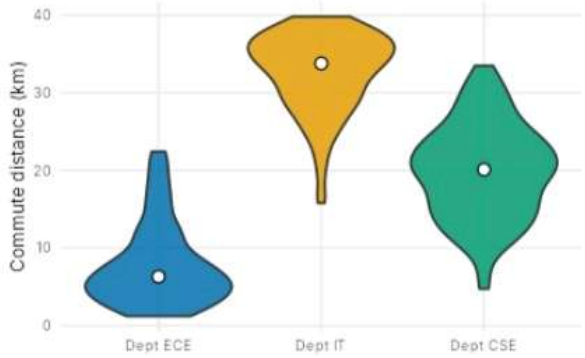
Violin	Skew (circle one)	Mean vs Median (circle)
Group 1	Right / Left / Symm.	Mean > / < / ~ Median
Group 4	Right / Left / Symm.	Mean > / < / ~ Median
Group 2	Right / Left / Symm.	Mean > / < / ~ Median

Chart 3: Resting heart rate (beats per min)



Violin	Skew (circle one)	Mean vs Median (circle)
Club D	Right / Left / Symm.	Mean > / < / ~ Median
Club B	Right / Left / Symm.	Mean > / < / ~ Median
Club E	Right / Left / Symm.	Mean > / < / ~ Median

Chart 4: Commute distance (km)



Violin	Skew (circle one)	Mean vs Median (circle)
Dept ECE	Right / Left / Symm.	Mean > / < / ~ Median
Dept IT	Right / Left / Symm.	Mean > / < / ~ Median
Dept CSE	Right / Left / Symm.	Mean > / < / ~ Median

Part B (2 marks)

B1. In Chart 1, violin "Store 4" is not symmetric. Towards which side is its tail, and which is the better measure of a typical order value here: mean or median? Give one reason. [1]

B2. Write one ggplot2 line that draws violins of value for each group from a data frame df (columns group, value), with the median shown. [1]

Activity - 4

A. Lokesh Kumar
1AIS24157.

Part A: Identify Skewness Using Violin Plots.

Chart 1: Order Value (Rs hundred)

Violin	Skew	Mean vs Median
Store 4	Left	Mean < Median
Store 2	Right	Mean > Median
Store 1	Left	Mean < Median.

Chart 2: Daily Steps (thousand steps).

Violin	Skew	Mean vs Median.
Group 1	Right	Mean > Median
Group 4	Symmetric	Mean ≈ Median
Group 2	Left	Mean < Median.

Chart 3: Resting Heart Rate

Violin	Skew	Mean vs Median.
Club D	Right	Mean > Median
Club B	Left	Mean < Median
Club E	Symmetric	Mean ≈ Median.

Chart 4: Commute Distance (km)

Violin	Skew	Mean Vs Median
Dept ECE	Right	Mean > Median
Dept IT	Left	Mean < Median
Dept CSE	Symmetric	Mean = Median

Part B: Answers.

B1. Chart 1 - Store 4

The tail extends towards the left. The median is a better measure of a typical order value because it is less affected by skewness.

B2. R code.

```
ggplot(df, aes(x=group, y=value)) + geom_violin() +
  stat_summary(fun = median, geom = "point")
```